Data & Analytics Consultant – Task Interview

Candidate Brief

Please complete the below technical task ahead of your interview. This is designed to help us understand how your technical skills and approach align to the expectations of the role.

There are two parts to this task:

1. Data extraction
2. Data science

We recommend spending no more than 2 hours on this in preparation for the interview. We understand it covers a broad range of topics and it might be tempting to spend longer, but please rest assured, we're not looking for a perfect answer.

Please submit both deliverables mentioned below at least 2 hours before your interview.

Part 1: Data Extraction Task

When we are deploying OneView, one of the initial requirements in the data acquisition process is to create procedures to extract data from the client's back end systems.

The data to be transferred to Xantura is first discussed, agreed and signed off after which an extract definition document is created and provided to the client.

While some clients undertake the data extraction themselves others require assistance and some opt to employ Xantura to complete the task in its entirety. For this test we will be assuming the latter.

Alongside this document you will have been provided with the following:

- A SQL script to create a simple example client database. (DACS_SETUP.sql)
- A SQL script to populate the database (DACS_INSERT.sql).
- An extract specification document 'DACS Test (DAVSV3) Specification.pdf'

Your task is as follows:

1. Create the example database
2. Analyse the tables and data in conjunction with the specification document identifying any problems (Formatting, Type mismatch, Ambiguity etc.) that must be addressed
3. Develop a SQL script to extract all the relevant data from the example DB in a format that satisfies all the requirements in the specification document

Deliverable: Please submit your script before your next interview where we will discuss your approach and also ask you to provide answers to the following questions:

- How could the script be altered to only extract deltas in subsequent runs?
- What options might the client be able to employ to run the extract on a scheduled basis?

Note: Assume Windows Server and MSSQL for all of the above.

Part 2: Data Science

Frailty Stratification for Elderly Population (65+)

Scenario: You are engaged by a UK local authority to develop a frailty stratification framework aimed at identifying individuals aged 65 and above who are at risk of becoming frail. The objective is to support early intervention, reduce hospital admissions, and improve quality of life through better resource planning and targeted outreach.

The local authority has access to various health and social care datasets, including general practice data, hospital records, demographic information, and social care service usage. They want a system that identifies individuals into Low, Medium, or High frailty risk.

Objective: Design core components of a predictive approach to stratify the frailty risk of individuals aged 65+ within a local authority's population, using available structured & unstructured data (such as case notes).

Your Task

Outline and explain your approach to the following:

1. **Problem Framing**

Describe how you would structure the entire solution:

- Build a framework and model architecture for feature engineering & model training
- List the types of data you would ideally want to use
- Suggest how frailty might be defined or labelled from that data

2. **Modelling Approach**

Provide a plan for developing a predictive model:

- How would you explore and understand the data?
- Outline a step-by-step plan for building a predictive model using structured & unstructured data
- Mention:
    - Key features you would consider
    - One or two model types you might use, and why
    - How you would evaluate model performance (name 2–3 appropriate metrics)

3. **Monitoring and Fine-Tuning**

- How would you track the model's performance after deployment?
- When would you decide to fine-tune or retrain the model?
- Name one way to keep the model fair and explainable.

Deliverables

Submit a short document or slide deck (max 2 pages or 5 slides) ahead of your interview, answering the above. Clear bullet points or short paragraphs are fine, depth of thinking is more important than format or polish.